COMPOSITE MATERIAL DATASHEETS
CFRP HTA/977-2-5HS

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MATERIAL IDENTIFICATION

Short informal name	HTA/977-2-5HS
Commercial name	Cytec Cycom 977-2A-37%-3KHTA-5H-280
Specifications	Airbus: ABS5003B-05 / AIMS05-01-004 (IPS 05-01-004-05)
Description	Carbon fibre reinforced prepreg. 5HS fabric / 280g/m2 / 180ºC curing class. Standard modulus fibre
Version	0.1 (29/11/2021) – Warning! Preliminary data valid only for trade-off studies

REFERENCES

- [Airbus] AIMS05-01-004 Is. 3 “Carbon fibre reinforced prepreg. 5HS fabric / 280g/m2 / 180ºC curing class. Standard modulus fibre. Material Specification.”
- [Airbus] X029RP1439435 Is. 10.3 “Summary of Design Values for Composite Materials”
- [Cytec] TECHNICAL DATA SHEET CYCOM® 977-2 and 977-2A PREPREG
- NT-L-ADP-04002 Is. D “Allowable pull-out load. Summary of test results for Z-19.775/Z-19.780 (AS4/8552)”

PHYSICAL PROPERTIES

Cured Ply Thickness (*CPT*)	0.280	mm
Density	1560	kg/m³
Tg-onset (wet)	130	ºC
Max. Operational Limit (*MOL*)	100	ºC

Coefficients of Thermal Expansion CTE (1/ºC)	Range	-55…20ºC	20…70ºC	70…120ºC
	α_1	2.3E-6	2.3E-6	2.3E-6
	α_2	2.3E-6	2.3E-6	2.3E-6
	α_3	3.4E-5	3.4E-5	3.4E-5

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
Values	62000	62000	4200	0.05	0.83

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13} -	ν_{23} -
Values					

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1, E_2 and G_{12} in buckling stability analyses
- Elastic modulus of the QI laminate is E_{QI} = 44469 MPa

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis	*E-KDF*			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{1t}	900	900	900	900	0.83	1.0	1.0	1.0	747	747	747	747
F_{1c}	900		650	545	0.83		0.722	0.606	747		540	452
F_{2t}	900	900	900	900	0.83	1.0	1.0	1.0	747	747	747	747
F_{2c}	900		650	545	0.83		0.722	0.606	747		540	452
F_{12}	119		82		0.83		0.689		98.8		68.1	

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (με)				B-basis	*E-KDF*			B-basis Values (με)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
ε_{tall}	11500	11500	11500	11500	0.83	1.0	1.0	1.0	9500	9500	9500	9500
ε_{call}	14500		10500	8800	0.83		0.722	0.606	12000		8700	7300

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis	*E-KDF*			B-basis Values			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{13} (MPa)	81		52		0.83		0.642		67.2		43.2	
F_{3t} (MPa)												
G_{IC} (N/mm)												
G_{IIC} (N/mm)												

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis	*E-KDF*			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
OHT_{QI}	275	275			0.83	1.000			228	228		
OHC_{QI}	290	290			0.83	1.000			241	241		
FHT_{QI}	275	275			0.83	1.000			228	228		
FHC_{QI}	370	370			0.83	1.000			307	307		
$F_{BR\text{-}QI}$	590	480		415	0.83	0.814		0.703	490	398		344

- For bolted joints analysis under in-plane bypass/bearing loads, before an official replacement is released, Airbus DS software URCO shall be used, selecting material “Group V Fabric”.
- For checking pull-out strength under tension in rivet/bolt use design curves in NT-L-ADP-04002-D

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and.

Property	B-basis Value for all conditions
TAI (με)	4100
CAI (με)	2900

SANDWICH WRINKLING

Parameters for wrinkling analysis of honeycomb sandwich panels with faces of this material (see SMM chapter 3510 “STRENGTH OF COMPOSITE SANDWICH PANELS”).

Configuration	C_1	C_2	C_3	C_4
Default CMH-17 values (conservative)	0.247	0.078	0.33	0.0